# Waterjet quote - 6061 plates

Four drawings, all **6061**: the **two Y front fins** nested in one square of 10 mm plate, the **two X gantry end plates**, the **three doubler blocks**, and the **X carriage plate**. **The material is supplied throughout** - every outline is the piece handed over, drawn to register against, and only the fins' two profiles are a cut. **On the fins the gap between the two tapered edges is called out on the drawing** - please say if it is too tight for your kerf. On the end plates the origin is on the part's factory long edge, not the sawn one. **The X carriage plate's sixteen ø5.2 holes are counterbored and that is not part of this job** - cut them through at 5.2 and the counterbores are done here. **Every drawing is a separate DXF** and the file name is printed on its page.

## What is in it

| File | Part | Material | Overall | Holes |
|---|---|---|---|---|
| `4-front-fins.dxf` | Y FRONT FIN - QTY 2 | STOCK 302 mm (11.890 in) x 302 mm (11.890 in) x 10 mm (0.394 in) 6061 | 302 × 302 mm | 8 × ø4.2 mm (0.165 in) · 20 × ø9 mm (0.354 in) |
| `5-x-end-plate-left.dxf` | X GANTRY END PLATE, LEFT-HAND - QTY 1 | | 245 × 154 mm | 16 × ø5.2 mm (0.205 in) · 4 × ø6 mm (0.236 in) · 2 × ø6.8 mm (0.268 in) · 4 × ø9 mm (0.354 in) · 1 × ø35 mm (1.378 in) |
| `6-x-end-plate-right.dxf` | X GANTRY END PLATE, RIGHT-HAND WITH THE TONGUE - QTY 1 | | 300 × 154 mm | 4 × ø4.2 mm (0.165 in) · 16 × ø5.2 mm (0.205 in) · 4 × ø6 mm (0.236 in) · 2 × ø6.8 mm (0.268 in) · 4 × ø9 mm (0.354 in) · 1 × ø35 mm (1.378 in) |
| `7-doublers.dxf` | DOUBLER BLOCKS - QTY 3 | | 154 × 163 mm | 12 × ø4.2 mm (0.165 in) |
| `8-x-carriage-plate.dxf` | X CARRIAGE PLATE - QTY 1 | STOCK 407 mm (16.024 in) x 154 mm (6.063 in) x 12.7 mm (0.500 in) 6061 | 407 × 154 mm | 24 × ø4.2 mm (0.165 in) · 16 × ø5.2 mm (0.205 in) · 4 × ø6 mm (0.236 in) · 1 × ø35 mm (1.378 in) |

## How to read the drawings

- **Units are millimetres** and the origin is the lower-left corner of every drawing, marked with a **red dot**. Inch equivalents are printed alongside for reference only — work to the millimetre figures.
- **Hole offsets from that origin are printed on each page** — the **X** values across the top, the **Y** values down the left side, each on a leader from its own tick. They are there to read the drawing by; the **DXF is the authority** and the offsets are taken from it.
- **R12 DXF, lines, circles and arcs only.** No polylines, no splines, no blocks. Outlines are closed.
- **Layers:** OUTLINE is the cut profile, HOLES are the holes, NOTES and PARTS are text and are not to be cut.
- **Where a drawing says STOCK, the material is supplied** and the outline drawn IS that piece — the job is the holes, and any excess falls outside them. Where it says MATERIAL, the shop supplies the plate and the outline is the part to cut.
- **Holes called out as edge-bored are not drawn.** They run into a part's end face rather than through it, and are drilled here.
- **Where two parts share one piece**, both are drawn in position on it with the kerf allowed between them.

## What would help on the quote

- Price **per drawing**, and separately the material where you are supplying it.
- Say if any hole is **below your minimum for the thickness** — a pilot that gets opened out here is fine, and so is a scribed mark, so tell us which you would rather cut.
- Note any outline you would rather nest differently. Nesting is yours to choose; the drawings do not assume one.

# Y FRONT FIN - QTY 2

STOCK 302 mm (11.890 in) x 302 mm (11.890 in) x 10 mm (0.394 in) 6061

UNITS mm - ORIGIN LOWER LEFT

BOTH PARTS NEST IN THE ONE SQUARE, HEAD TO TAIL. THE TAPER IS ONE SHARED CUT - NOT TWO

CUT ORDER: HOLES FIRST, THEN BOTH PERIMETERS, THEN THE SHARED DIAGONAL LAST

THE DIAGONAL IS THE CUT THAT PARTS THEM - UNTIL IT RUNS THE PLATE IS ONE PIECE AND NOTHING CAN SHIFT

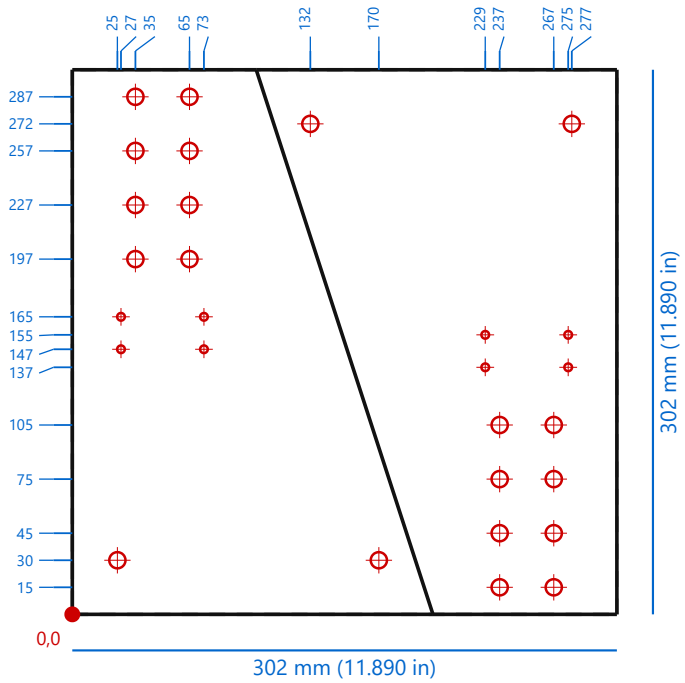
THE KERF COMES OFF BOTH PARTS AT THE TAPER. THAT IS ACCEPTED - DO NOT ADD A WEB BETWEEN THEM

THE SQUARE IS NOMINAL. INDEX EACH PART'S WIDE BASE EDGE ON A REAL PLATE EDGE, OPPOSITE EDGES

ANY SHORTFALL IN THE PLATE COMES OFF THE TAPERED ENDS, NOT OFF A BASE EDGE

**Holes** 8 × ø4.2 mm (0.165 in) · 20 × ø9 mm (0.354 in)

Drawn at 1:4 from the DXF itself · 4-front-fins.dxf · offsets are HOLE CENTRES from the red 0,0 dot - X across the top, Y down the left side



# X GANTRY END PLATE, LEFT-HAND - QTY 1

PART 245 mm (9.646 in) x 154 mm (6.063 in) x 12.19 mm (0.480 in) 6061 - ALREADY CUT TO SIZE

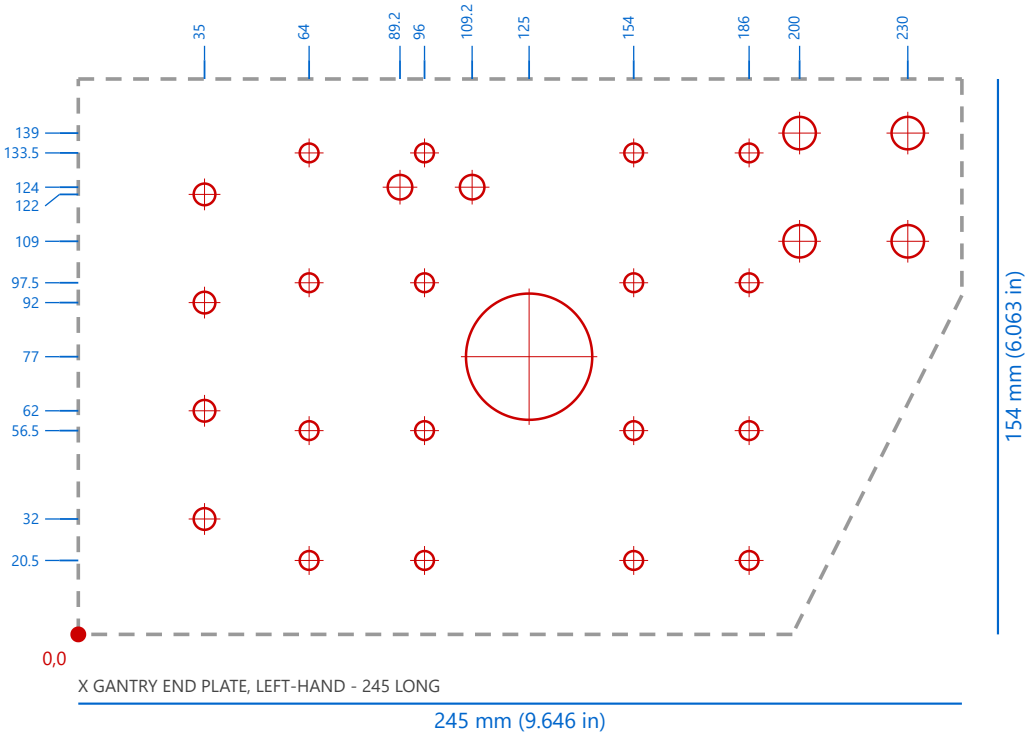
UNITS mm - ORIGIN LOWER LEFT - THE OUTLINE IS REFERENCE, NOT A CUT LINE

ORIGIN IS ON THE FACTORY LONG EDGE - the UNCUT one, NOT the sawn edge the doubler came off

4 EDGE-BORED HOLE(S) ARE NOT DRAWN - they run into an end face and are drilled here

**Holes** 16 × ø5.2 mm (0.205 in) · 4 × ø6 mm (0.236 in) · 2 × ø6.8 mm (0.268 in) · 4 × ø9 mm (0.354 in) · 1 × ø35 mm (1.378 in)

Drawn at 1:2 from the DXF itself · 5-x-end-plate-left.dxf · offsets are HOLE CENTRES from the red 0,0 dot - X across the top, Y down the left side



# X GANTRY END PLATE, RIGHT-HAND WITH THE TONGUE - QTY 1

PART 300 mm (11.811 in) x 154 mm (6.063 in) x 12.19 mm (0.480 in) 6061 - ALREADY CUT TO SIZE

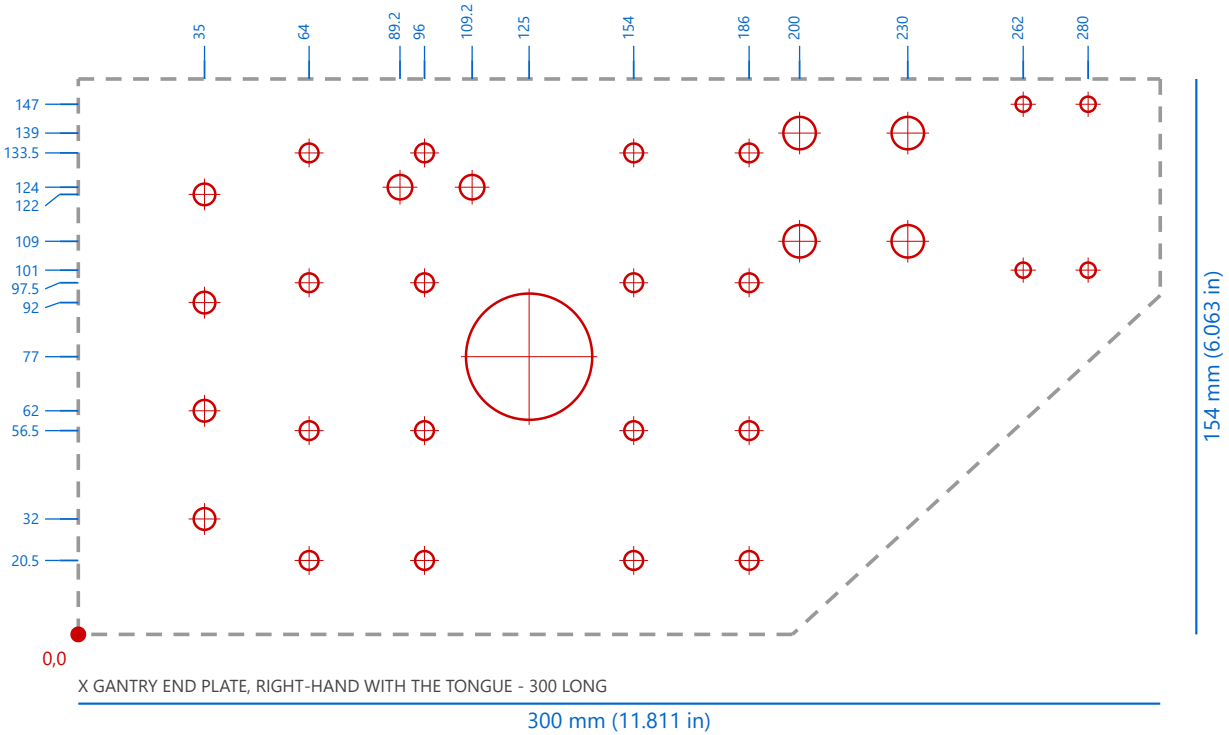
UNITS mm - ORIGIN LOWER LEFT - THE OUTLINE IS REFERENCE, NOT A CUT LINE

ORIGIN IS ON THE FACTORY LONG EDGE - the UNCUT one, NOT the sawn edge the doubler came off

4 EDGE-BORED HOLE(S) ARE NOT DRAWN - they run into an end face and are drilled here

**Holes** 4 × ø4.2 mm (0.165 in) · 16 × ø5.2 mm (0.205 in) · 4 × ø6 mm (0.236 in) · 2 × ø6.8 mm (0.268 in) · 4 × ø9 mm (0.354 in) · 1 × ø35 mm (1.378 in)

Drawn at 1:2 from the DXF itself · 6-x-end-plate-right.dxf · offsets are HOLE CENTRES from the red 0,0 dot - X across the top, Y down the left side



# DOUBLER BLOCKS - QTY 3

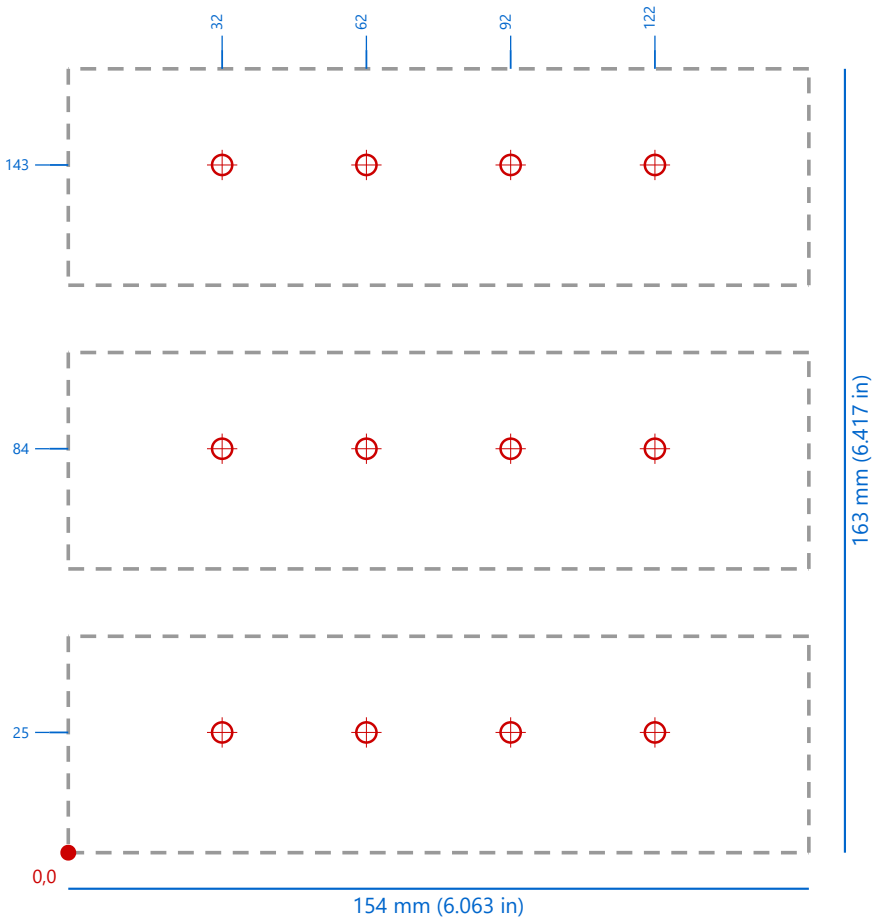
PART 154 mm (6.063 in) x 45 mm (1.772 in) x 12.19 mm (0.480 in) 6061 - ALREADY SAWN TO SIZE

UNITS mm - ORIGIN LOWER LEFT OF EACH BLOCK - THE OUTLINE IS REFERENCE, NOT A CUT LINE

4 EDGE-BORED HOLE(S) PER BLOCK ARE NOT DRAWN - they run into an end face and are drilled there

**Holes** 12 × ø4.2 mm (0.165 in)

Drawn at 1:1.5 from the DXF itself · 7-doublers.dxf · offsets are HOLE CENTRES from the red 0,0 dot - X across the top, Y down the left side



# X CARRIAGE PLATE - QTY 1

STOCK 407 mm (16.024 in) x 154 mm (6.063 in) x 12.7 mm (0.500 in) 6061

UNITS mm - ORIGIN LOWER LEFT

**Holes** 24 × ø4.2 mm (0.165 in) · 16 × ø5.2 mm (0.205 in) · 4 × ø6 mm (0.236 in) · 1 × ø35 mm (1.378 in)

Drawn at 1:2 from the DXF itself · 8-x-carriage-plate.dxf · offsets are HOLE CENTRES from the red 0,0 dot - X across the top, Y down the left side

